

Design Rules Summary

for Drawing Layers

2026/06/02

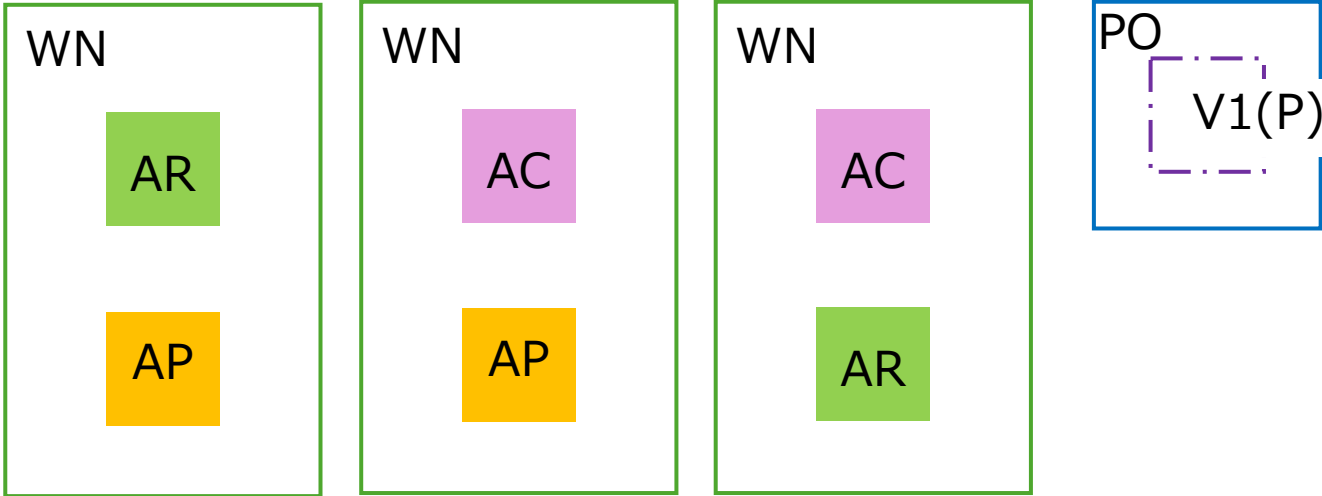
Reflected GitHub issues

<https://github.com/OpenSUSI/TR-1um/issues>

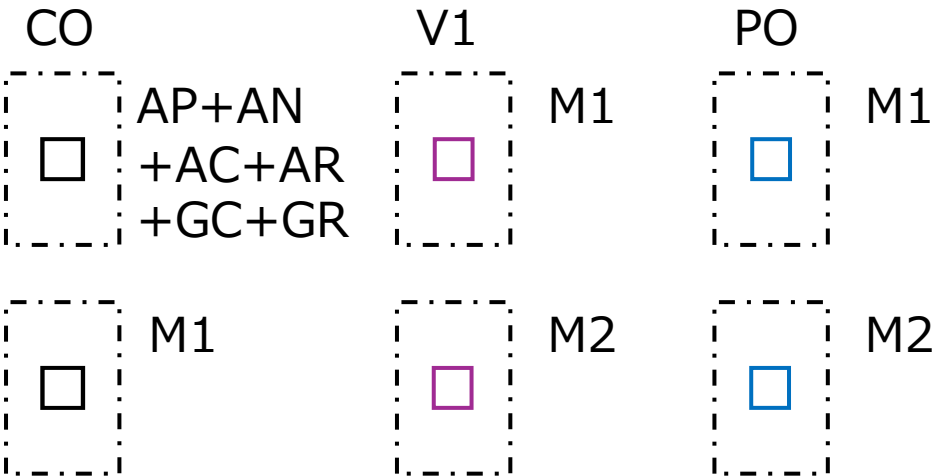
Drawing Layer Table

	Original MASK layers				Proposed Drawing layers					
	Name	NUM	TYPE		Name	NUM	TYPE			
1	PSUB	140	0		WN	140	0			
2	NW	36	0					MDP generatioed		
3	HVNW	141	0					MDP generatioed		
4	L	3	0					MDP generatioed		
					AP	3	1	Active for P+		
					AN	3	2	Active for N+		
					AR	3	3	Active for resistor		
					AC	3	4	Active for capacitance		
5	NF	25	0					MDP generatioed		
6	PF	26	0					MDP generatioed		
7	CL	143	0					MDP generatioed		
8	HPBE	144	0					No USE		
9	HNBE	145	0					No USE		
10	PBE	146	0					MDP generatioed		
11	NBE	147	0					MDP generatioed		
12	HPM	33	0					MDP generatioed		
13	RHP	148	0					No USE		
14	SG	8	0							
					GC	8	1	Gate Poly transistor		
					GR	8	2	Gate POLY resistor		
15	PM	35	0					MDP generatioed		
16	NM	7	0					MDP generatioed		
17	R	12	0					MDP generatioed		
18	PSD	9	0					MDP generatioed		
19	NSD	28	0					MDP generatioed		
20	CONT	11	0		CO	11	0			
21	M1	13	0		M1	13	0			
22	TC	19	0		V1	19	0			
23	M2	20	0		M2	20	0			
24	PRO	14	0		PO	14	0			
	TXM1	48	0		TXM1	48	0	M1 Label		
	PIN	48	1		PIN	48	1	M1 Pin		
	TXM2	49	0		TXM2	49	0	M2 Label		
	MASK	63	0		MASK	63	0	Waiver region		
	PTECT	63	1		PTECT	63	1	Protect region (LOGO)		
	SCRB	80	0		SCRB	80	0	Waiver region (EDGE SEAL)		
	prBoundary	235	0		prBoundary	235	0	Cell boundary		

Cat-1: Prohibited



Rule	L1	L2	Func	MIN	MAX
WN.Z1	WN(R)	AP	None	0.0	
WN.Z2	WN(C)	AP	None	0.0	
WN.Z3	WN(C)	AR	None	0.0	
CO.Z1	CO	AA+GC+GR	None	0.0	
CO.Z3	CO	M1	None	0.0	
V1.Z1	V1	M1	None	0.0	
V1.Z2	V1	M2	None	0.0	
PO.Z1	PO	M1	None	0.0	
PO.Z2	PO	M2	None	0.0	
PO.Z3	PO	V1(P)	Exist	0.0	

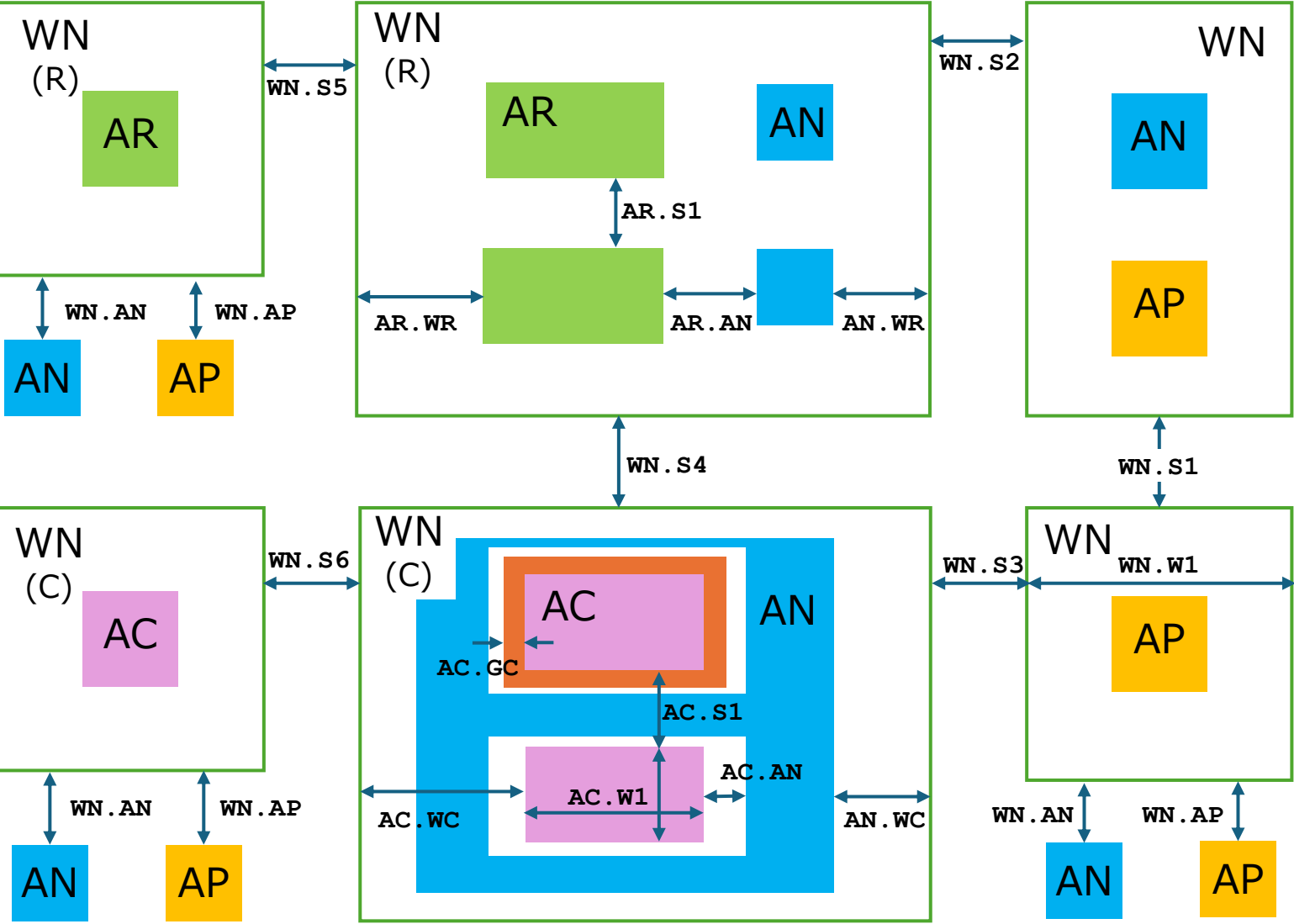


Function:

Wmin/max: L1 Width min/max
Smin/max: L1 Space min/max (L1=L2)
Smin/max: L1 Separate L2 min/max
Emin/max: L1 Enclosed L2 min/max
Fmin/max: L1 Enclosing L2 min/max
ECmin: MOS Endcap min
Wfix: L1 Windth equal
Sfix: Space equal
Donut: L1 Souuounded by L2
None: Not Exist

V1(P) : V1 on PE

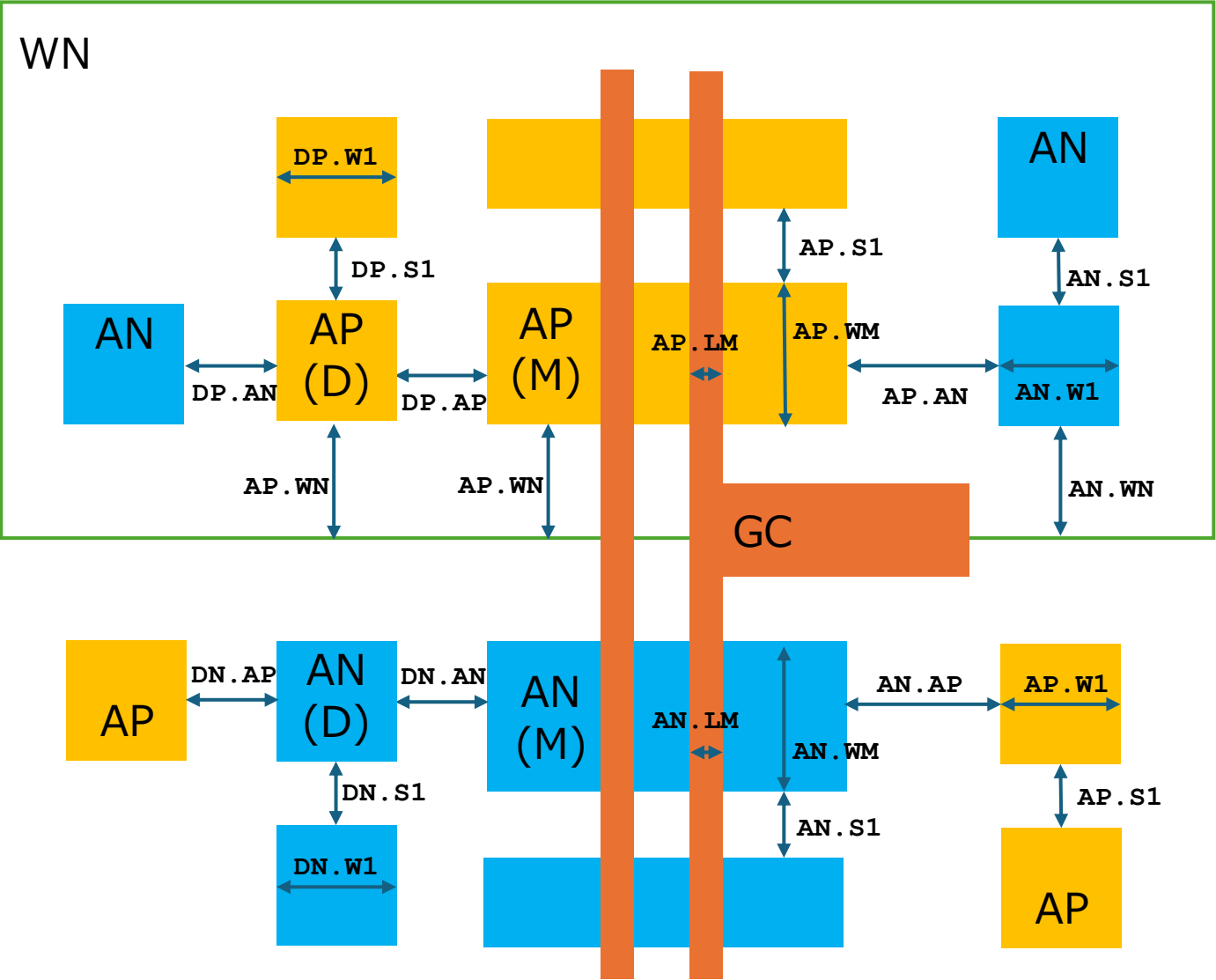
Cat-2: WN to AP/AN/AR/AC related



Rule	L1	L2	Func	MIN	MAX
WN.W1	WN		Wmin	8.0	
WN.S1	WN	WN	Smin	12.0	
WN.S2	WN	WN(R)	Smin	9.5	
WN.S3	WN	WN(C)	Smin	12.0	
WN.S4	WN(R)	WN(C)	Smin	9.5	
WN.S5	WN(R)	WN(R)	Smin	8.0	
WN.S6	WN(C)	WN(C)	Smin	12.0	
WN.AP	WN	AP	Smin	5.0	
WN.AN	WN	AN	Smin	10.0	
AR.S1	AR	AR	Smin	4.0	
AR.WR	AR	WN(R)	Emin	10.0	
AN.WR	AN	WN(R)	Emin	5.0	
AR.AN	AR	AN	Smin	4.0	
AC.W1	AC		Wmin/max	28.5	120.0
AC.S1	AC	AC	Smin	???	???
AC.WC	AC	WN(C)	Emin	???	???
AN.WC	AN	WN(C)	Emin	3.0	
AC.AN	AC	AN	Sfix	2.8	2.8
AC.GC	AC	GC	Efix	1.4	1.4
AC.R1	AC	AN(C)	Donut	0.0	

WN(R) : WN overlapped AR
WN(C) : WN overlapped AC
AN(C) : AN in WN(C)

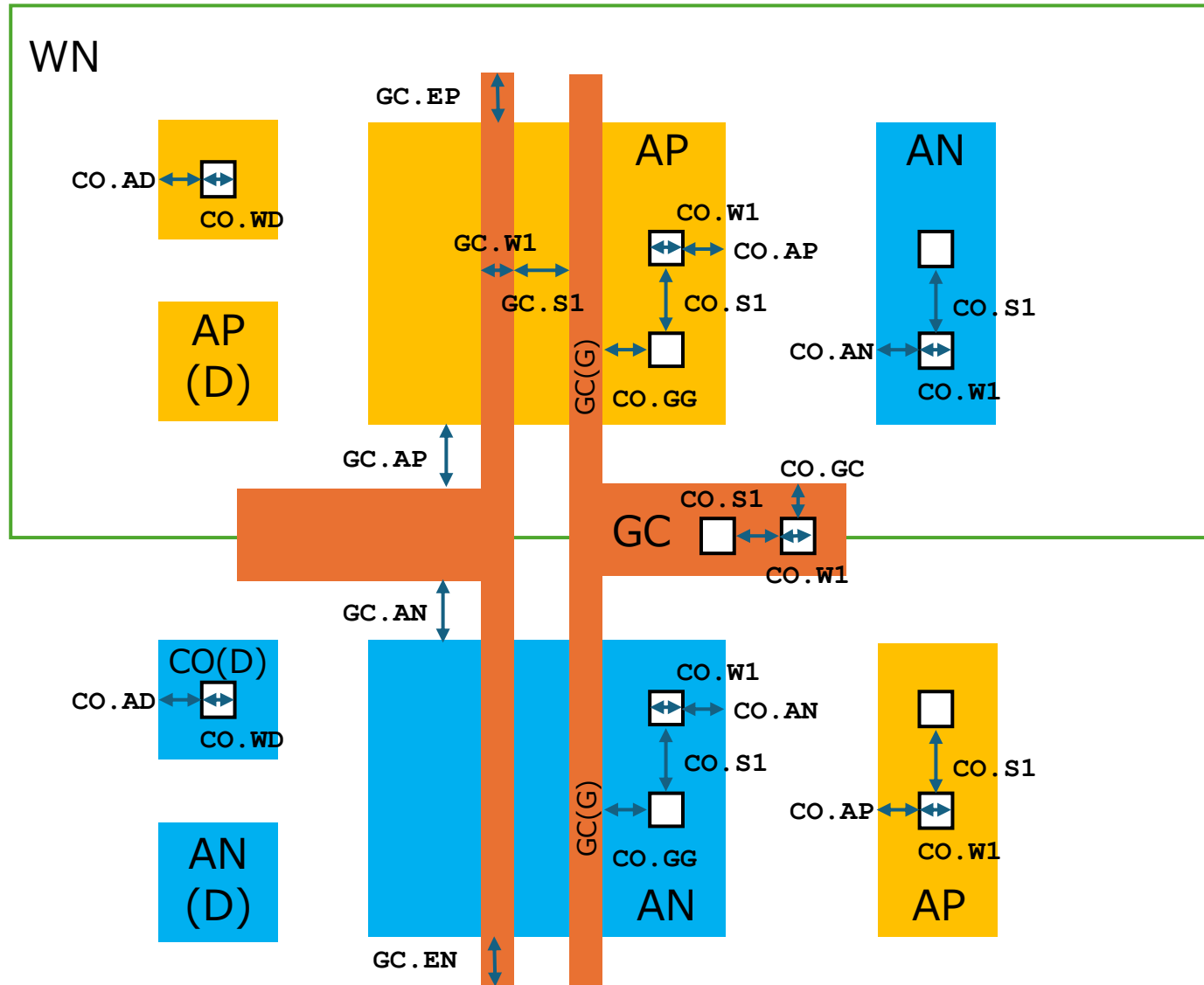
Cat-3: MOS transistor related



Rule	L1	L2	Func	MIN	MAX
AP.W1	AP		Wmin	1.4	
AP.S1	AP	AP	Smin	1.4	
AP.AN	AP	AN	Smin	2.8	
AP.WM	PMOS	AP-GC	Wmin/max	3.4	60.0
AP.LM	PMOS	GC-AN	Lmin/max	1.0	30.0
AP.WN	AP	WN	Emin	7.0	
DP.W1	DP		Wmin	2.8	
DP.S1	DP	DP	Smin	1.4	
DP.AP	DP	AP(M)	Smin	2.8	
DP.AN	DP	AN	Smin	2.8	
AN.W1	AN		Wmin	1.4	
AN.S1	AN	AN	Smin	1.4	
AN.AP	AN	AP	Smin	2.8	
AN.WM	NMOS	AN-GC	Wmin/max	3.4	60.0
AN.LM	NMOS	GC-AN	Lmin/max	1.0	30.0
AN.WN	AN	WN(M)	Emin	5.0	
DN.W1	DN		Wmin	2.8	
DN.S1	DN	DN	Smin	1.4	
DN.AP	DN	AP	Smin	2.8	
DN.AN	DN	AN(M)	Smin	2.8	

AN (M) : AN-DN
AP (M) : AP-DP
WN (M) : WN overlapped AN/AP

Cat-4: Contact and Poly related



Rule	L1	L2	Func	MIN	MAX
GC.W1	GC		Wmin	1.0	
GC.S1	GC	GC	Smin	1.2	
GC.AP	GC	AP	Smin	0.4	
GC.AN	GC	AN	Smin	0.4	
GC.EP	PMOS	GC	ECmin	1.2	
GC.EN	NMOS	GC	ECmin	1.2	
CO.W1	CO		Wfix	1.0	1.0
CO.S1	CO	CO	Smin	1.0	
CO.WD	CO(D)		Wfix	1.2	1.2
CO.AD	CO(D)	AP+AN	Emin	1.2	
CO.AP	CO	AP	Emin	0.8	
CO.AN	CO	AN	Emin	0.8	
CO.GG	CO	GC(G)	Smin	1.0	
CO.GC	CO	GC+GR	Emin	0.8	

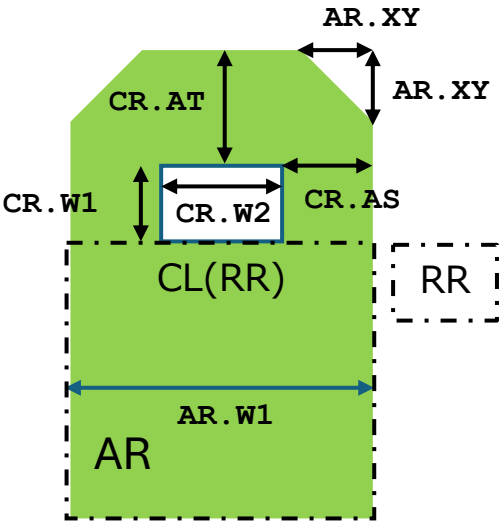
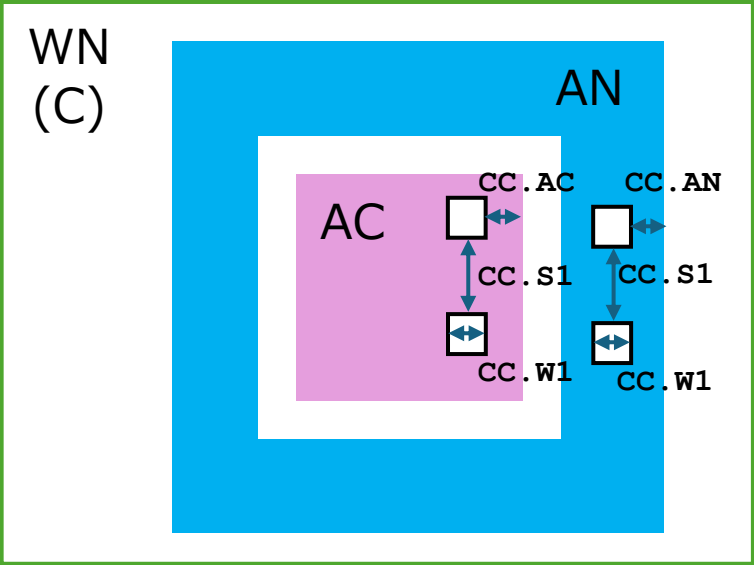
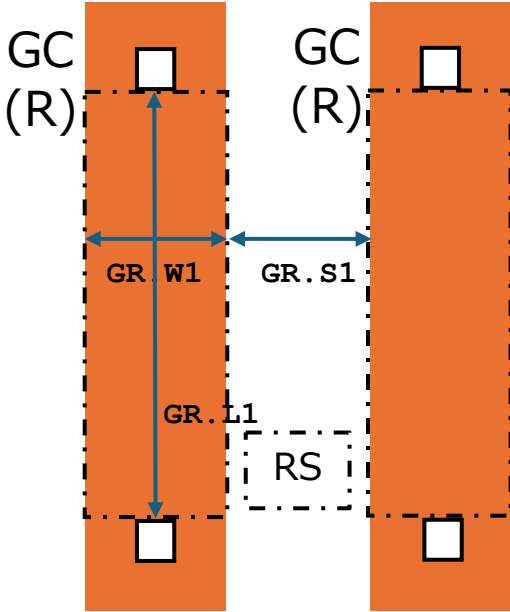
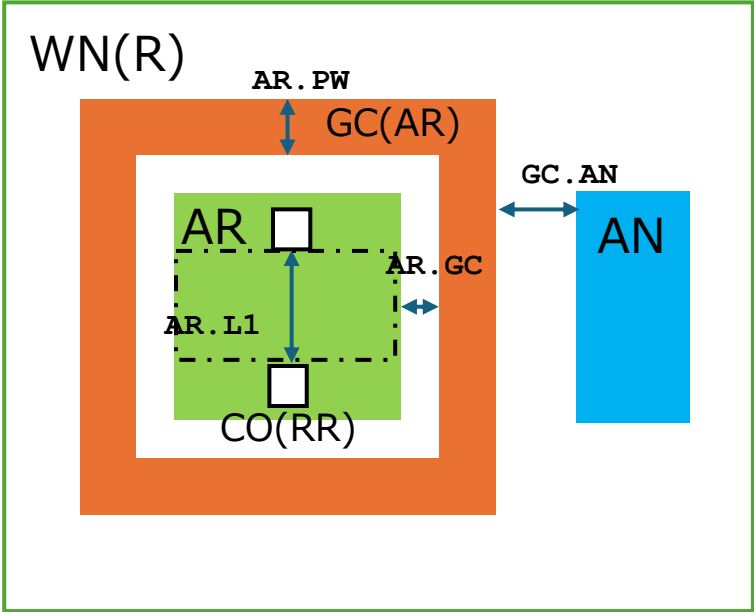
AP (D) : AP not interacted GC

AN (D) : AN not interacted GC

$$\text{CO}(D) : \text{CO on AP}(D) / \text{AN}(D)$$

GC (G) : GC for PMOS/NMOS

Cat-5: Resistor(AR/RS) and Capacitor(AC) related

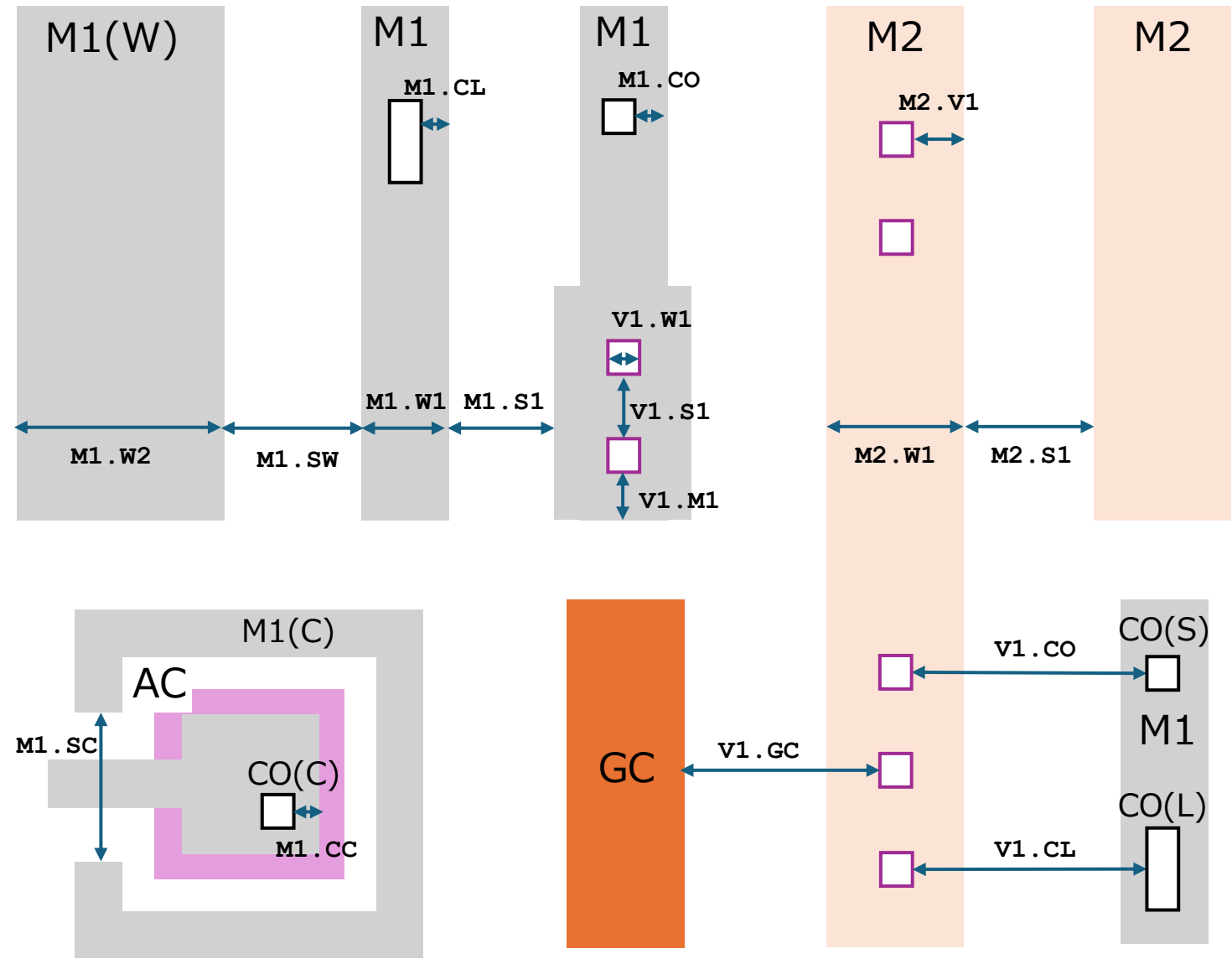


R.Effective region

Rule	L1	L2	Func	MIN	MAX
AR.W1	RR	RR(L)	Wmin/max	2.8	20.0
AR.L1	RR	RR(W)	Lmin/max	13.0	100.0
AR.GC	GC	AR	Smin	1.0	
AR.PW	GC(R)		Wmin	2.0	
AR.XY	AR	Bevel	XYmin	1.0	
GC.AN	GC(R)	AN	Smin	???	???
GC.R1	AR	GC(R)	Donut	0.0	
GC.R2	GC(R)	WN(R)	TieDown	0.0	
GR.W1	RS	RS(L)	Wmin/max	4.0	20.0
GR.L1	RS	RS(W)	Lmin/max	20.0	100.0
GR.S1	RS	RS	Smin	2.0	
CR.W1	CO(RR)		Wmin	1.0	
CR.W2	CL(RR)		Wmin/max	1.0	97.5
CR.AT	CO(RR)	AR(T)	Emin	1.5	
CR.AS	CO(RR)	AR(S)	Emin	1.0	
CC.W1	CO(C)		Wfix	1.2	1.2
CC.S1	CO(C)	CO(C)	Sfix	1.2	1.2
CC.AC	CO(C)	AC	Emin	2.9	
CC.AN	CO(C)	AN	Emin	1.2	

GC(AR) : GC in WN(R)
RS(L) : Length of AR effective
RS(W) : Width of AR effective
CO(RR) : CO short side on AR
CL(RR) : CO large side on AR
AR(T) : AR top fringe
AR(S) : AR side fringe
AR.L1 : allow only 4,6,12,20
CO(C) : CO on AC

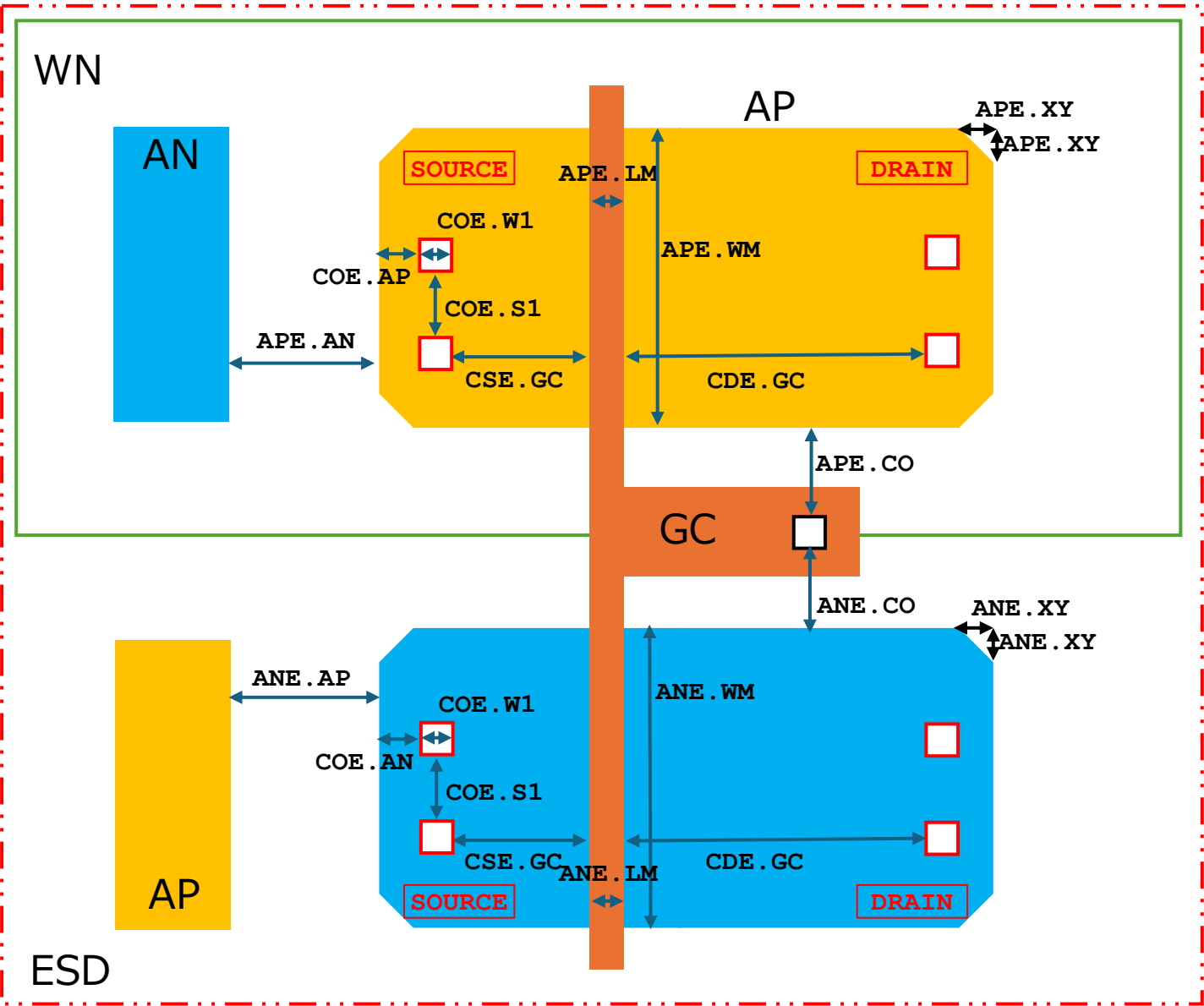
Cat-6: M1 /V1 /M2 related



Rule	L1	L2	Func	MIN	MAX
M1.W1	M1		Wmin/max	1.8	10.0
M1.W2	M1(W)		Wmin	10.0	
M1.S1	M1	M1	Smin	1.4	
M1.SC	M1(C)	M1(C)	Sfix	10.0	10.0
M1.CO	M1	CO(S)	Fmin	0.8	
M1.CC	M1	CO(C)	Fmin	1.3	
M1.CL	M1	CO(L)	Fmin	1.2	
M1.SW	M1(W)	M1	Smin	2.0	
V1.W1	V1		Wfix	1.4	1.4
V1.S1	V1	V1	Smin	1.5	
V1.M1	V1	M1	Emin	1.0	
V1.GC	V1	GC	Smin	1.2	
V1.CO	V1	CO	Smin	1.0	
V1.CL	V1	CO(L)	Smin	1.4	
M2.W1	M2		Wmin	3.0	
M2.S1	M2	M2	Smin	2.0	
M2.V1	M2	V1	Fmin	1.0	

M1 (C) : M1 in WN (C)
CO (C) : CO on AC
CO (S) : CO small contact
CO (L) : CO large contact (only on AR)
M1 (W) : Wide M1

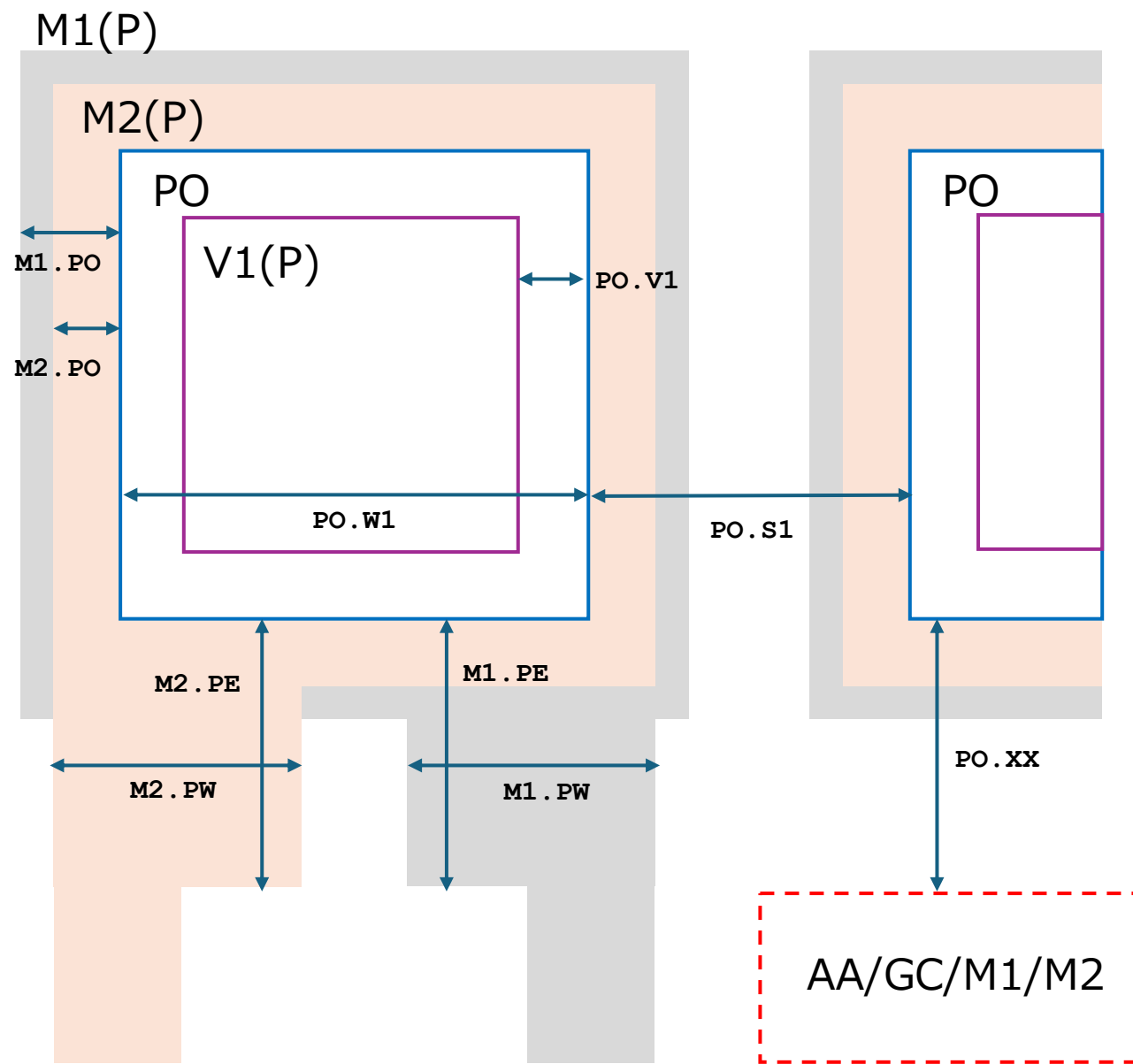
Cat-7: ESD MOS transistor related



Rule	L1	L2	Func	MIN	MAX
APE.WM	PMOSE	APE-GC	Wmin/max	11.0	60.0
APE.LM	PMOSE	GC-ANE	Lmin/max	2.0	30.0
APE.AN	APE	AN	Smin	10.0	
APE.XY	APE	Bevel	XYmin	1.4	
APE.CO	APE	CO	Smin	2.0	
ANE.WM	NMOSE	ANE-GC	Wmin/max	11.0	60.0
ANE.LM	NMOSE	GC-ANE	Lmin/max	2.0	30.0
ANE.AP	ANE	AP	Smin	10.0	
ANE.XY	ANE	Bevel	XYmin	1.4	
ANE.CO	ANE	CO	Smin	2.0	
COE.W1	CO(E)		Wfix	3.0	3.0
COE.S1	CO(E)	CO(E)	Smin	1.6	
CDE.GC	CD(E)	GC	Smin	7.0	
CSE.GC	CS(E)	GC	Smin	3.0	
COE.AP	CO(E)	AP	Emin	4.0	
COE.AN	CO(E)	AN	Emin	4.0	

APE/ANE) : AP/AN on ESD)
CO (E) : CO on ESD

Cat-8: PAD related



Rule	L1	L2	Func	MIN	MAX
PO.W1	PO		Wmin	70.0	
PO.S1	PO		Smin	64.0	
M2.PO	M2 (P)	PO	Fmin	5.0	
M1.PO	M1 (P)	PO	Fmin	5.0	
PO.V1	PO	V1 (P)	Emin	10.0	
PO.AP	PO	AP	Smin	14.0	
PO.AN	PO	AN	Smin	14.0	
PO.GC	PO	GC	Smin	14.0	
PO.M1	PO	M1	Smin	14.0	
PO.M2	PO	M2	Smin	14.0	
M2.PE	M2	Extention		19.0	
M2.PW	M2	Extention	Wmin	40.0	
M1.PE	M2	Extention		19.0	
M1.PW	M2	Extention	Wmin	40.0	

** Not yet implemented

M1 (P) : M1 on PO
M2 (P) : M2 on PO
V1 (P) : V1 on PO

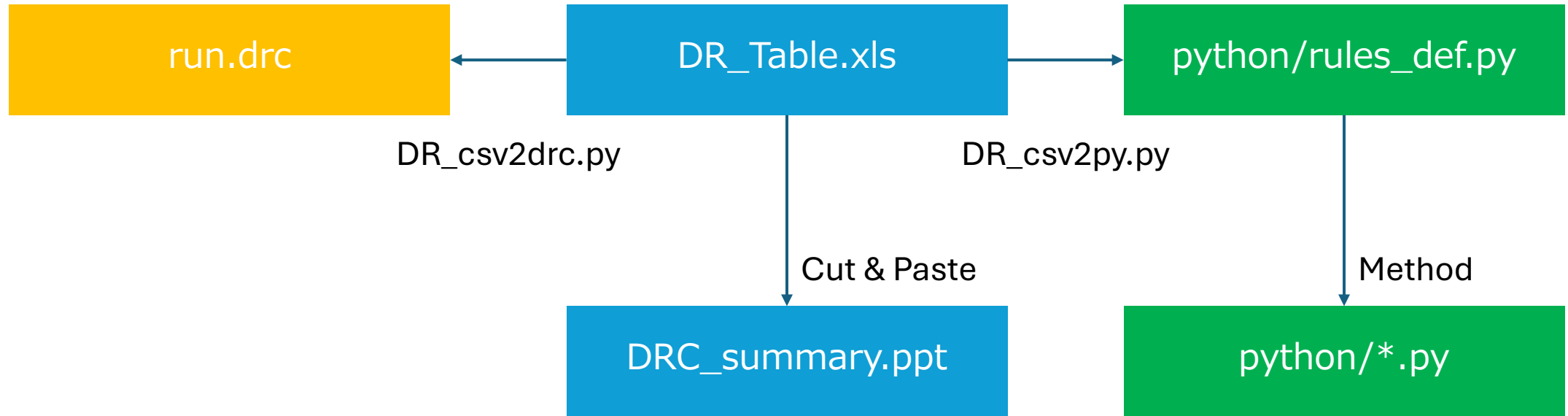
Design Rule control

DRC integrity

DRC

Document

PCell



ALL Critical Dimension parameters are referenced by 'XX.YY' rule's name.

Design to Tapeout Process

Mask Data Preparation

Designer

local



[push]

remote



[action]
Precheck
DRC
MDP



[clone]



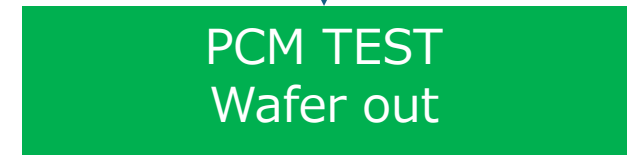
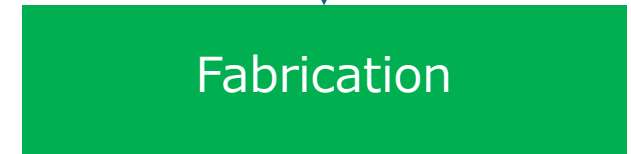
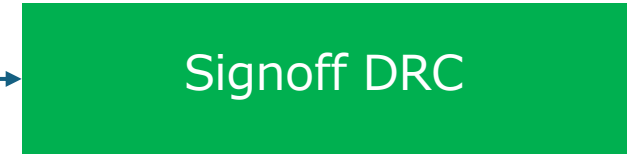
[Import artifact]



[action]
Logo insertion
Aggregation
DRC



Tokai Rika



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